Munshi Mahbubur Rahman

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RESEARCH INTERESTS

AI/ML Model Evaluation & Validation · Algorithmic Fairness · Uncertainty & Robustness Characterization · Trustworthy AI · Data-Centric Machine Learning

SUMMARY

AI/ML researcher (Ph.D. candidate) with 5+ years developing and rigorously evaluating machine learning models in high-stakes domains, including healthcare. Core strength is building evaluation methodology for AI systems: characterizing model performance and uncertainty, quantifying bias and robustness under noisy or limited ground truth, and designing reproducible pipelines that support transparent, defensible conclusions — directly applicable to establishing scientific validity for AI/ML in forensic pattern-evidence analysis.

EDUCATION

Ph.D. in Information Systems, University of Maryland, Baltimore County (GPA 3.80/4.00) *Expected Dec. 2026*

M.S. in Information Systems, University of Maryland, Baltimore County (GPA 3.80/4.00) *Aug. 2023*

B.Sc. in Electronics and Comm. Engineering, Khulna University of Engineering & Technology (GPA 3.41/4.00) *May 2016*

RESEARCH EXPERIENCE

The Latent Lab, UMBC *Baltimore, MD*

Graduate Research Assistant *2020–Present*

- Conduct NSF-funded research (IIS-1927486, IIS-2046381) on fairness-aware ML with direct relevance to rigorous evaluation of AI systems in high-stakes decision settings.
- Developed a unified evaluation framework for AI/ML models spanning multiple, sometimes competing performance criteria, enabling transparent characterization of trade-offs and operational limitations — the same class of problem as validating model output against multiple forensic decision criteria.
- Built reproducible Python/PyTorch pipelines for model training and evaluation, including subgroup performance assessment, sensitivity analysis, and custom metric design, to support rigorous, defensible model documentation.
- Designed and validated a dynamic weighting algorithm (LA-FW) for combining multiple evaluation metrics efficiently, benchmarked against exhaustive search across three datasets (healthcare, criminal-justice, income-prediction) — direct experience benchmarking AI/ML approaches against baseline procedural methods.
- Applied methods to healthcare utilization prediction (MEPS) and other benchmark datasets; analyzed model behavior under class imbalance, low-prevalence outcomes, and constrained/noisy data regimes.
- Earlier research (2019–2022) included end-to-end modeling for fake news detection using Hugging Face transformer-based language models, validated under noisy labels and limited ground truth — methodology directly relevant to forensic datasets where confirmed ground truth is scarce.

SAIL Lab, UMBC *Baltimore, MD*

Graduate Research Assistant *2018–2019*

- Conducted applied ML experiments on multimodal physiological and behavioral sensor data, performing signal preprocessing, exploratory analysis, and model evaluation for pattern-recognition tasks in human-centered sensing.

PUBLICATIONS

Peer-Reviewed

- Rahman, M.M., Pan, S., & Foulds, J.R. (2026). When Fairness Metrics Help Each Other: Look-Ahead Optimization for Multi-Metric Fairness Weighting. ACM GoodIT 2026. Accepted.
- Rahman, M.M., Pan, S., & Foulds, J.R. (2024). Towards a Unifying Human-Centered AI Fairness Framework. ACM GoodIT 2024. Best Work-in-Progress Paper Award.

Preprints

- Rahman, M.M., Pan, S., & Foulds, J.R. (2025). A Unifying Human-Centered AI Fairness Framework. arXiv preprint.

Earlier Scholarly Publications

- Rahman, M.M., & Foulds, J.R. (2020). End-to-End Joint Modeling for Fake News Detection. UMBC Student Collection.

SELECTED PROJECTS

Healthcare Utilization Prediction (MEPS) — Subgroup Reliability Analysis *2024–Present*

- Analyzed model performance and fairness under class imbalance and low-prevalence outcomes; evaluated subgroup-level disparities and stability across training regimes.

KEY TECHNICAL SKILLS

Programming: Python, C, C++, SQL; working knowledge of Java, MATLAB

Machine Learning: Neural networks, probabilistic classifiers, fairness-aware ML methods, traditional ML and transformer-based NLP models

Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Hugging Face Transformers

Model Evaluation & Validation: Uncertainty and sensitivity analysis, subgroup/reliability analysis, bias and robustness auditing (AIF360), custom evaluation-metric design, benchmarking against baseline methods

Data: Sensor and healthcare/public-sector datasets; class imbalance and low-prevalence outcome handling

Tools: Docker, Git, Jupyter, TensorBoard

AWARDS

- Best Work-in-Progress Paper Award, ACM International Conference on Information Technology for Social Good (GoodIT 2024), Bremen, Germany.

PRIOR INDUSTRY EXPERIENCE

Robi Axiata Ltd. — Project Coordinator, Core Network Planning *Dhaka, Bangladesh | Feb 2018 – Apr 2018*

- Monitored and validated core network performance metrics (GGSN, SGSN) against capacity and traffic targets, flagging anomalies for planning decisions.

National Credit & Commerce Bank Ltd. — Database Administrator *Dhaka, Bangladesh | Sep 2016 – Aug 2017*

- Validated and monitored transactional database systems for data integrity, access control, and regulatory compliance in a financial environment.

TEACHING EXPERIENCE

Department of Information Systems, UMBC — Course Co-Developer *Summer 2024*

- Co-developed online AI course materials (Introduction to Artificial Intelligence, UMBC–edX MicroMasters) emphasizing evaluation, responsible AI, and societal impacts — experience translating technical AI/ML concepts for varied audiences.

UMBC — Graduate Teaching Assistant *2019–2023*

- Supported instruction for IS 804 (Quantitative Methods), IS 450 (Networks), and IS 320 (Business Applications), including labs, grading, and student assistance.

CERTIFICATIONS

- ASEE DELTA Future Faculty Institute
- CCNA (Cisco Certified Network Associate)